Midterm
March 18th

1. (20 points) A store sells a perishable product. Suppose that the daily demand D for the product is i.i.d with the following distribution.

d	15	20	30	40
$\mathbb{P}(D=d)$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{3}$

Unused items from one day *cannot* be used in the following day. Suppose that each item sells at \$120. Each item costs \$20 to order, and each leftover item has to be disposed at a cost of \$40. The store makes a daily order at the end of each day for the product to arrive at the beginning of next day (before sales commence).

   (a) (8 points) Find the optimal order quantity to maximize the expected profit per day.

   (b) (5 points) Compute the expected profit per day, given that the optimal order quantity from part (a) is ordered.

   (c) (4 points) Compute the 95% confidence interval of the cumulative profit in 100 days, given that the optimal order quantity from part (a) is ordered. (leaving your answer as an expression is fine.)

   (d) (3 points) Consider now there is a fixed cost to make an order. The fixed cost is \$150 if the order quantity is positive but less than 40, and \$100 if the order quantiy is at least 40. How many items should it order to maximize the expected profit for the following day? Show your work to support your claim.

2. (30 points) Consider a DTMC with state space $\{a, b, c, d, e\}$ and transition matrix

$$P = \begin{bmatrix} 0 & 0 & 0.8 & 0.2 & 0 \\ 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0.6 & 0 & 0 & 0.4 \end{bmatrix}.$$

   (a) (4 points) What is $\mathbb{P}(X_{28} = b | X_{26} = e)$?

   (b) (4 points) What is $\mathbb{P}(X_2 = c, X_4 = e | X_1 = b)$?

   (c) (5 points) Identify all the recurrent classes of the DTMC.

   (d) (5 points) What is the period of each state?

   (e) (4 points) Find approximately $\mathbb{P}(X_{100} = b | X_0 = c)$.

   (f) (4 points) Find approximately $\mathbb{P}(X_{100} = e | X_0 = a)$.

   (g) (4 points) Compute $\mathbb{E}_c(T_c)$, the expected time for the Markov chain to first return to state c.

3. (15 points) A two-side reflected random walk $\{X_n\}$ on $[0, N]$ is defined as

$$P_{i,i+1} = 1 - P_{i,i-1} = p, \text{ for } 1 \le i \le N-1; P_{01} = 1 - P_{00} = p; P_{N,N} = 1 - P_{N,N-1} = p.$$

Consider $N = 3$ and $p = 2/3$.

   (a) (5 points) Find a stationary distribution of the DTMC.

   (b) (5 points) Let $T_0 = \min\{n \ge 1 | X_n = 0\}$ be the first time to reach state 0. Compute $\mathbb{E}_2[T_0]$, the expected number of steps to reach 0 starting from 2.

   (c) (5 points) Let $T_0^{(2)} = \min\{n > T_0 | X_n = 0\}$ be the second time to reach state 0. Compute $\mathbb{E}_2[T_0^{(2)}]$.

4. (30 points) Tom drinks a bottle of yogurt after breakfast and dinner, both with probability 0.5, unless there is no yogurt in the fridge. Every morning *before breakfast*, his mother will check the fridge and go to shop. If there is no yogurt in the fridge, she will buy 3 bottles of yogurt and put them in the fridge *at noon*. Let X_n denote the number of bottles of yogurt at the beginning of day n (before breakfast) and D_n denote the amount of yogurt Tom drinks on day n. Consider that the initial quantity of yoghurt does not exceed 3 bottles.

   (a) (8 points) What is the distribution of D_n when $X_n = 0, 1, 2$ and 3, repsectively?

   (b) (10 points) Verify that $\{X_n\}$ is a DTMC by writing X_{n+1} as a function of X_n and U_{n+1}, where $\{U_n\}$ is an i.i.d. sequence. Determine the transition probabilities of the DTMC $\{X_n\}$. (Hint: U_n can be a random vector.)

   (c) (8 points) Suppose it costs 5 dollars to buy a bottle of yogurt. What is the (long run) expected amount of money his mother spends on yogurt every day?

   (d) (4 points) Define $Y_n = 1$ when $X_n \geq 2$, reflecting adequate supply of yogurt for day n, and $Y_n = 0$ otherwise. Is $\{Y_n\}$ a DTMC? Prove your claim.

5. (5 points) Suppose $\{X_n\}$ is an irreducible DTMC with state space S and transition matrix P. Consider another DTMC $\{Y_n\}$ with transition matrix Q defined as $Q_{ij} = \frac{1_{\{i=j\}}+P_{ij}}{2}$ for all $i, j \in S$. Here $1_{\{i=j\}}$ is 1 if $i = j$ and 0 otherwise. Prove that $\lim_{k\to\infty} (Q^k)_{ij} = \pi(j)$, where π is a stationary distribution of $\{X_n\}$ and Q^k is k-step transition matrix of $\{Y_n\}$.